

MAT1002 Lecture 10,

Outline

- Cross products (Continued)
- Lines and Planes in $\mathbb{R}^3$
- Cylinders and quadric surfaces

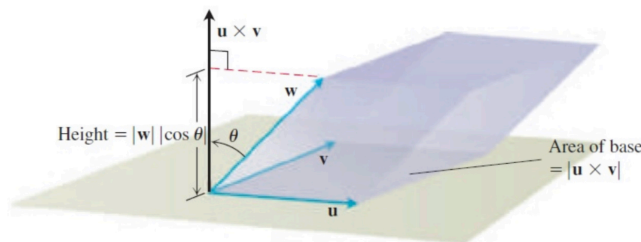
Cross Products

Triple Scalar Products : $(\vec{u} \times \vec{v}) \cdot \vec{w}$

By a property of dot product, if θ is angle between $\vec{u} \times \vec{v}$ and $\vec{w}$, then

$$(\vec{u} \times \vec{v}) \cdot \vec{w} = |\vec{u} \times \vec{v}| |\vec{w}| \cos \theta,$$

from which we can see that the absolute value of $(\vec{u} \times \vec{v}) \cdot \vec{w}$ is the volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$ and $\vec{w}$.



Determinant Form of $(\vec{u} \times \vec{v}) \cdot \vec{w}$

It can be checked by direct computation that

$$(\vec{u} \times \vec{v}) \cdot \vec{w} = \det \begin{bmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix}.$$

Fact about determinant : Swapping two rows in a matrix changes the sign of its determinant.

By applying row swapping operation twice in the matrix above, we have

$$\det \begin{bmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix} = \det \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}.$$

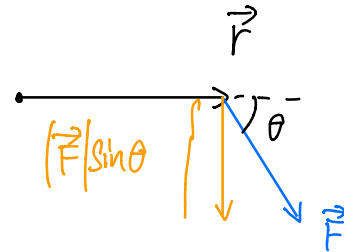
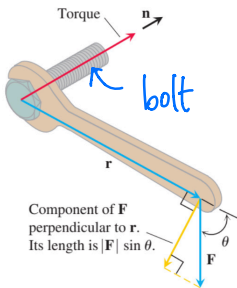
Hence

$$(\vec{u} \times \vec{v}) \cdot \vec{w} = \det \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}.$$

Version in the book.

Torque

A force $\vec{F}$ is applied to a wrench to turn the bolt, as indicated below.



The **torque** is a force (vector) whose:

- magnitude measures the tendency of rotation of the wrench.
- direction is perpendicular to the plane of rotation.

It is defined by $\vec{\tau} := \vec{r} \times \vec{F}$. Note that

$$|\vec{\tau}| = |\vec{r} \times \vec{F}| = |\vec{r}| |\vec{F}| \sin \theta.$$

“tau”

Lines in $\mathbb{R}^3$

A **vector equation** for the line L through a point $\underbrace{(x_0, y_0, z_0)}_{P_0}$ parallel to $\vec{v}$ is

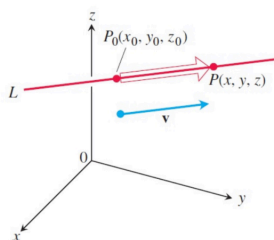
$$\vec{r}(t) = \vec{r}_0 + t\vec{v}, \quad t \in \mathbb{R},$$

where $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$. The corresponding **parametric equations** for L is

position vector $\vec{OP}_0$ of the point P_0 .

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3, \quad t \in \mathbb{R},$$

where $\vec{v} = \langle v_1, v_2, v_3 \rangle$.



Example

The line through $(5, -2, 0)$ and $(-1, 3, 2)$ has parametric equations

$$x = 5 + 6t, \quad y = -2 - 5t, \quad z = -2t, \quad t \in \mathbb{R}.$$

Remark: • A line has many different vector equations

$$\vec{r}(t) = \vec{r}_0 + t\vec{v},$$

Since there are different choices for $\vec{r}_0$ and $\vec{v}$.

• $\vec{r}$ can be viewed as a function $\vec{r}: \mathbb{R} \rightarrow \mathbb{R}^3$,

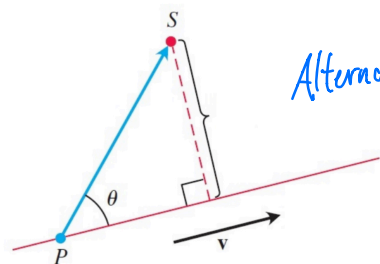
$$\vec{r}(t) := \langle x_0 + tv_1, y_0 + tv_2, z_0 + tv_3 \rangle.$$

Position vector of points on the line.

Distance from a Point to a Line in the Space

Let L be a line (in the space) through a point P parallel to $\vec{v}$, and let S be a point. Then the distance d from S to L can be computed by

$$d = \frac{|\vec{PS} \times \vec{v}|}{|\vec{v}|}.$$



Alternatively, $d = |\vec{PS} - \text{proj}_{\vec{v}} \vec{PS}|$

Planes in $\mathbb{R}^3$

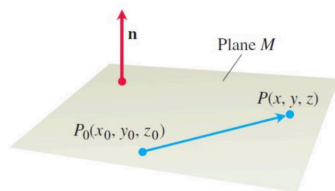
Each plane M in the space has a vector $\vec{n}$ that is perpendicular to the plane; such vector is called a **normal vector** to the plane. If $P_0(x_0, y_0, z_0)$ is a point in M , then M consists of points $P(x, y, z)$ such that

$$\overrightarrow{P_0P} \cdot \vec{n} = 0.$$

In particular, if $\vec{n} = \langle A, B, C \rangle$, then

(Component) Equation of a plane

$$M = \{(x, y, z) : A(x - x_0) + B(y - y_0) + C(z - z_0) = 0\}.$$



It can also be described by

$$\vec{n} \cdot \vec{OP} = \vec{n} \cdot \vec{OP}_0$$

$$Ax + By + Cz = D,$$

where $D = Ax_0 + By_0 + Cz_0$ and $\langle A, B, C \rangle$ is a normal vector of the plane.

$$\vec{n} \cdot \vec{OP}_0$$

$$\vec{n}$$

Example

Find an equation of the plane passing through the points $P(1, 3, 2)$, $Q(3, -1, 6)$ and $R(5, 2, 0)$.

Sol (Outline): $\vec{PQ} \times \vec{PR} = \langle 12, 20, 14 \rangle$.

← Can pick this as $\vec{n}$.

$\vec{n} \cdot \vec{OP} = 100$ ($= \vec{n} \cdot \vec{OQ} = \vec{n} \cdot \vec{OR}$)

Equation: $12x + 20y + 14z = 100$.

Distance from a Point to a Plane

Let P be a point on a plane with a normal vector $\vec{n}$, and let S be any point. By considering vector projection (Page 17), the distance from S to the plane is

$$\left| \frac{\vec{PS} \cdot \vec{n}}{|\vec{n}|} \right|.$$

Scalar component of $\vec{PS}$ in the direction of $\vec{n}$.

Q: How to pick a point on the plane $Ax + By + Cz = D$?

e.g. Find the distance from $S(1, 1, 3)$ to M given by $3x + 2y + 6z = 6$.

Sol (Outline): $P(0, 0, 1)$ is on M , and $\vec{PS} = \langle 1, 1, 2 \rangle$.

$\vec{n} = \langle 3, 2, 6 \rangle$.

Distance = $\left| \frac{\vec{PS} \cdot \vec{n}}{|\vec{n}|} \right| = \frac{17}{7}$.

The intersection of two nonparallel planes is a line.

Exercise

Find parametric equations for the line which is the intersection of the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Sol (Outline):

- Any vector $\perp$ to both normals

$$\vec{n}_1 = \langle 3, -6, -2 \rangle \quad \text{and} \quad \vec{n}_2 = \langle 2, 1, -2 \rangle.$$

is parallel to the line.

- $\vec{n}_1 \times \vec{n}_2 = \langle 14, 2, 15 \rangle.$

- Find a point on the line (how?): $(10, 0, \frac{15}{2})$ is on the line.

- $x = 10 + 14t, \quad y = 2t, \quad z = \frac{15}{2} + 15t.$

More Terminology

Def:

- The **angle** between two planes is the angle θ , with $\theta \in [0, \frac{\pi}{2}]$, made by their normal vectors
- Two planes are **parallel** if their normal vectors are parallel.
- Two lines (in $\mathbb{R}^3$) are **skew** if they do not intersect and are not parallel.

e.g. Given lines $L_1: x=1+t, y=-2+3t, z=4-t$
and $L_2: x=2s, y=3+s, z=-3+4s$:

(a) Show that L_1 and L_2 are skew.

(b) Find the shortest distance between them.

Sol: (a). Their direction vectors $\vec{v}_1 = \langle 1, 3, -1 \rangle$ and $\vec{v}_2 = \langle 2, 1, 4 \rangle$
are not parallel, so $L_1 \nparallel L_2$.

$$\begin{aligned} \cdot \text{ Solve } \begin{cases} 1+t=2s & \textcircled{1} \\ -2+3t=3+s & \textcircled{2} \\ 4-t=-3+4s & \textcircled{3} \end{cases} \quad \textcircled{1} \& \textcircled{2} \Rightarrow t=\frac{11}{5} \& s=\frac{8}{5}. \\ \text{ But they do not satisfy } \textcircled{3} \end{aligned}$$

So system has no solution, and L_1 does not intersect L_2

• So L_1 and L_2 are skew.

(b). L_1 and L_2 must be contained in two parallel planes M_1 and M_2 ,
with $M_1 \neq M_2$. The shortest distance between L_1 and L_2 equals the
distance between M_1 and M_2 .

• A common normal of M_1 and M_2 is given by

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \langle 13, -6, -5 \rangle.$$

• Distance d between M_1 and M_2 can be found by finding a
point P_1 on M_1 then finding distance between P_1 and M_2 .

• Set $t=0 \Rightarrow P_1 = (1, -2, 4)$.

• Set $s=0 \Rightarrow P_2 = (0, 3, -3)$ is on M_z .

• Knowing P_2 and $\vec{n}$, M is given by

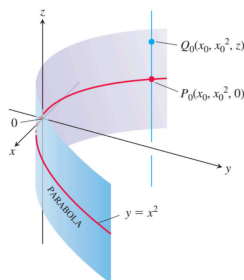
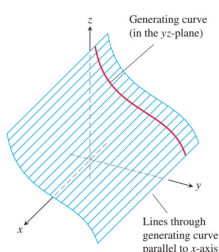
$$13x - 6(y-3) - 5(z+3) = 0, \text{ or } 13x - 6y - 5z = -3.$$

• Now

$$d = \left| \frac{\vec{P_2 P_1} \cdot \vec{n}}{|\vec{n}|} \right| = \left| \frac{\langle 1, -5, 7 \rangle \cdot \langle 13, -6, -5 \rangle}{(169 + 26 + 25)^{\frac{1}{2}}} \right| = \frac{8}{\sqrt{230}}.$$

Cylinders and Quadric Surfaces

Def: A **cylinder** is a surface that consists of all lines that parallel to a given line and pass through a given plane curve.



Def: A **quadric surface** is the set of all points (x, y, z) in $\mathbb{R}^3$ that satisfy a degree-two equation $f(x, y, z) = 0$.

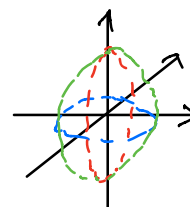
We will investigate those given by the equations

$$Ax^2 + By^2 + Cz^2 + D = 0 \quad \text{or} \quad Ax^2 + By^2 + Cz = 0.$$

e.g. The equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ describe the **ellipsoid**.

By setting $x=0$, $y=0$ or $z=0$, one at a time, we can see that its intersections with the coordinate planes are ellipses.

If $a=b=c$, then it is a sphere.



e.g. The **hyperbolic paraboloid** is described by

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}.$$

$$x=0 \Rightarrow \frac{y^2}{b^2} c = z : \text{parabola}$$

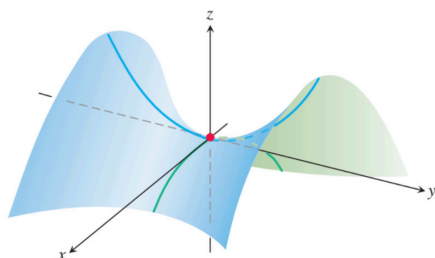
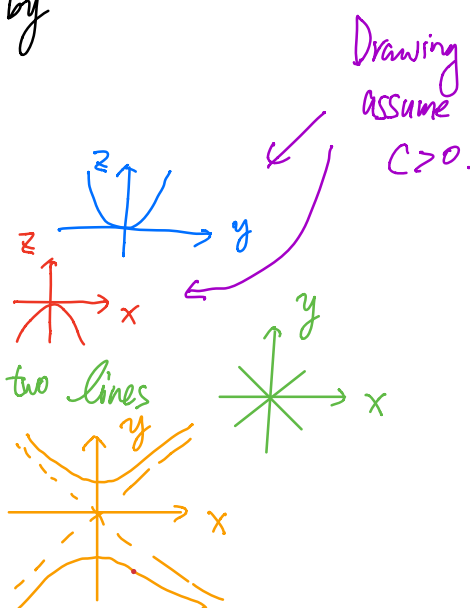
$$y=0 \Rightarrow -\frac{x^2}{a^2} c = z : \text{parabola}$$

$$z=0 \Rightarrow \left(\frac{y}{b}\right)^2 = \left(\frac{x}{a}\right)^2 \Rightarrow y = \pm \frac{b}{a} x : \text{two lines}$$

$$z=c \Rightarrow \frac{y^2}{b^2} - \frac{x^2}{a^2} = 1 : \text{hyperbola}$$

(Similar for $z=z_0$)

Ex: Use Geogebra to plot such a surface.



e.g. The elliptic paraboloid is described by
(or elliptical)

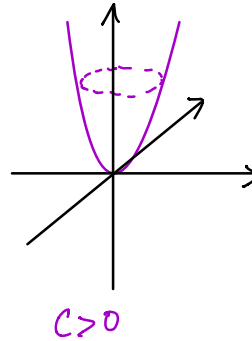
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$$

$$x=0 \Rightarrow \frac{y^2}{b^2} c = z : \text{parabola}$$

$$y=0 \Rightarrow \frac{x^2}{a^2} c = z : \text{parabola}$$

$$z=0 \Rightarrow (x,y,z) = (0,0,0) : \text{origin}$$

$$z=z_0 > 0 : \text{ellipse.} \\ (\text{if } c > 0)$$



e.g. The elliptic cone is described by
(or elliptical)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$$

$$x=0 : \text{two lines}$$

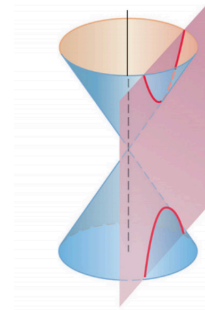
$$y=0 : \text{two lines}$$

$$x=x_0 \neq 0 : \text{hyperbola}$$

$$y=y_0 \neq 0 : \text{hyperbola}$$

$$z=0 : (x,y,z) = (0,0,0)$$

$$z=z_0 \neq 0 : \text{ellipse}$$



See the book for more quadric surfaces.